

PostgreSQL Tablespace

PostgreSQL Tablespace Management

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0. Introduction

What is a Tablespace?

--> Tablespace is a logical structure in which objects are stored.

--> A tablespace is simply a physical storage location on disk where PostgreSQL stores database objects (tables, indexes,

etc.).

--> By default database objects are stored in the current default tablespace of the database.

List tablespaces:

```
\db+ or select * from pg_tablespace;
```

Types of Tablespaces: Default Tablespaces and Non-Default Tablespaces

I. Default Tablespaces

These come built-in with PostgreSQL:

`pg_global` - for shared system data

`pg_default` - for regular user data (tables, indexes, etc.)

FEATURE	PG_DEFAULT	PG_GLOBAL
Purpose	Stores regular user data	Stores global system data
Used by default?	Yes, for tables/indexes	No, only used internally
Can store user tables?	✓ Yes	✗ No
Location	\$PGDATA/base/	\$PGDATA/global/
Droppable?	✗ No	✗ No

II. Non-Default Tablespaces :

--> These are created by users to store data in custom locations.

--> Useful for managing storage better (e.g., putting large tables on faster disks).

1. How big can a PostgreSQL tablespace grow?

There is no fixed maximum size for a tablespace in PostgreSQL itself.

The limit depends on your operating system and filesystem where the tablespace directory is located.

For example:

On modern filesystems like ext4 or XFS, a single file can be several terabytes (TB) or even petabytes (PB).

PostgreSQL breaks large tables into multiple files (each typically up to 1 GB) internally, so extremely large tablespaces are supported.

Your disk/storage capacity is the main practical limit.

2. Create a Tablespace

```
[root@pg17 ~]# mkdir -p /pgData/pgsql15/edbua_tbs01
[root@pg17 ~]# mkdir -p /pgData/pgsql15/edbua_tbs02
[root@pg17 ~]# mkdir -p /pgData/pgsql15/edbua_ixtbs01
[root@pg17 ~]# mkdir -p /pgData/pgsql15/edbua_temptbs01
[root@pg17 ~]# chown postgres:postgres /pgData/pgsql15/edbua*
[root@pg17 ~]# chmod 700 /pgData/pgsql15/edbua*
[root@pg17 ~]#
```

```
[root@pg17 ~]# su - postgres
```

```
[postgres@pg17 ~]$ psql
```

```
psql (15.13)
```

```
Type "help" for help.
```

```
postgres=# CREATE TABLESPACE edbua_tbs01 LOCATION
'/pgData/pgsql15/edbua_tbs01';
```

```
CREATE TABLESPACE
```

```
postgres=# CREATE TABLESPACE edbua_tbs02 LOCATION
'/pgData/pgsql15/edbua_tbs02';
```

```
CREATE TABLESPACE
```

```
postgres=# CREATE TABLESPACE edbua_ixtbs01 LOCATION
'/pgData/pgsql15/edbua_ixtbs01';
```

```
CREATE TABLESPACE
postgres=#
```

3. Create a Database with default tablespace

```
postgres=# CREATE DATABASE edbua TABLESPACE edbua_tbs01;
CREATE DATABASE
postgres=#
postgres=# SELECT oid, datname, dattablespace, datistemplate,
datdba FROM pg_database WHERE datname = 'edbua';
   oid   | datname | dattablespace | datistemplate | datdba
-----+-----+-----+-----+-----
 92666   | edbua   |               | f              |      10
(1 row)

postgres=#
```

4. Create Tables on Data tablespace

```
postgres=# \c edbua
You are now connected to database "edbua" as user "postgres".
edbua=# CREATE TABLE emp01 (id int, name text) TABLESPACE
edbua_tbs01;
CREATE TABLE
edbua=# CREATE TABLE emp02 (id int, name text) TABLESPACE
edbua_tbs01;
CREATE TABLE
edbua=# CREATE TABLE emp03 (id int, name text) TABLESPACE
edbua_tbs01;
CREATE TABLE
edbua=#
```

5. Create Index on Index tablespaces

```
edbua=# CREATE INDEX emp01_ix ON emp01(name) TABLESPACE
edbua_ixtbs01;
```

```

CREATE INDEX
edbua=# CREATE INDEX emp02_ix ON emp02(name) TABLESPACE
edbua_ixtbs01;
CREATE INDEX
edbua=# CREATE INDEX emp03_ix ON emp03(name) TABLESPACE
edbua_ixtbs01;
CREATE INDEX
edbua=# SELECT c.oid, c.relname, c.relkind, t.spcname AS
tablespace, c.relfilenode
edbua-# FROM pg_class c
edbua-# LEFT JOIN pg_tablespace t ON c.reltablespace = t.oid
edbua-# WHERE c.relname IN
('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');
   oid | relname   | relkind | tablespace   | relfilenode
-----+-----+-----+-----+-----
 92685 | emp01     | r       |              |          92685
 92690 | emp02     | r       |              |          92690
 92695 | emp03     | r       |              |          92695
 92700 | emp01_ix  | i       | edbua_ixtbs01 |          92700
 92701 | emp02_ix  | i       | edbua_ixtbs01 |          92701
 92702 | emp03_ix  | i       | edbua_ixtbs01 |          92702
(6 rows)

edbua=#

```

If a table or index is created in the database's default tablespace, PostgreSQL does not explicitly record the tablespace in pg_class.reltablespace.

But when it's created in a non-default tablespace, PostgreSQL records that tablespace OID in pg_class.reltablespace.

If you want to always see the tablespace name (even when it's the default)

```

edbua=# SELECT
edbua-#      c.oid,
edbua-#      c.relname,
edbua-#      c.relkind,
edbua-#      COALESCE(t1.spcname, t2.spcname) AS tablespace,
edbua-#      c.relfilenode
edbua-# FROM pg_class c
edbua-# LEFT JOIN pg_tablespace t1 ON c.reltablespace = t1.oid

```

```

edbua=# LEFT JOIN pg_database d ON d.datname =
current_database()
edbua=# LEFT JOIN pg_tablespace t2 ON d.dattablespace = t2.oid
edbua=# WHERE c.relname IN
('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');
   oid | relname | relkind | tablespace | relfilenode
-----+-----+-----+-----+-----
 92685 | emp01   | r       | edbua_tbs01 |          92685
 92690 | emp02   | r       | edbua_tbs01 |          92690
 92695 | emp03   | r       | edbua_tbs01 |          92695
 92700 | emp01_ix | i       | edbua_ixtbs01 |          92700
 92701 | emp02_ix | i       | edbua_ixtbs01 |          92701
 92702 | emp03_ix | i       | edbua_ixtbs01 |          92702
(6 rows)

edbua=#

```

pg_tblspc/<tbs_oid>/PG_15_<catalog_version>/<db_oid>/<relfilenode>

```

edbua=# SELECT relname, pg_relation_filepath(oid) AS filepath
edbua=# FROM pg_class
edbua=# WHERE relname IN
('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');
 relname |          filepath
-----+-----
 emp01   | pg_tblspc/92662/PG_15_202209061/92666/92685
 emp02   | pg_tblspc/92662/PG_15_202209061/92666/92690
 emp03   | pg_tblspc/92662/PG_15_202209061/92666/92695
 emp01_ix | pg_tblspc/92664/PG_15_202209061/92666/92700
 emp02_ix | pg_tblspc/92664/PG_15_202209061/92666/92701
 emp03_ix | pg_tblspc/92664/PG_15_202209061/92666/92702
(6 rows)

edbua=#

```

6. Create a Dedicated Tablespace for Temp

```

postgres=# CREATE TABLESPACE edbua temptbs01 LOCATION
'/pgData/pgsql15/edbua_temptbs01';

```

```
CREATE TABLESPACE
postgres=#
```

7. Assign Temp Tablespace to a Database

```
postgres=# ALTER DATABASE edbua SET temp_tablespaces =
edbua_temptbs01;
ALTER DATABASE
postgres=#

postgres=# \c edbua
You are now connected to database "edbua" as user "postgres".
edbua=#

edbua=# CREATE TEMP TABLE emp_tmp (id int, name text);
CREATE TABLE
edbua=#

edbua=# SELECT c.oid, c.relname, t.spcname,
pg_relation_filepath(c.oid)
edbua-# FROM pg_class c
edbua-# LEFT JOIN pg_tablespace t ON c.reltablespace = t.oid
edbua-# WHERE c.relname = 'emp_tmp';
   oid | relname |      spcname      |
pg_relation_filepath
-----+-----+-----+-----+-----
-----
 92705 | emp_tmp | edbua_temptbs01 |
pg_tblspc/92665/PG_15_202209061/92666/t4_92705
(1 row)

edbua=#

pg_tblspc/<tbs_oid>/PG_15_<catalog_version>/<db_oid>/<relfilen
ode>
```

8. Check Tablespace OIDs

```
edbua=# SELECT oid, spcname AS tablespace_name,
pg_tablespace_location(oid) AS location FROM pg_tablespace
```

```
ORDER BY oid;
```

oid	tablespace_name	location
1663	pg_default	
1664	pg_global	
24581	dell_ts_data	/u01/data/dell_ts_data
24582	dell_ts_index	/u01/data/dell_ts_index
24594	dell_tbs	/pgData/pgsql15/data
24597	trd_tb01	/pgData/pgsql15/data/trd_tb01
24598	trd_ix01	/pgData/pgsql15/data/trd_ix01
32787	geb_tb	/pgData/pgsql15/pgtb/geb_tb
32788	geb_ix	/pgData/pgsql15/pgix/geb_ix
92662	edbua_tbs01	/pgData/pgsql15/edbua_tbs01
92663	edbua_tbs02	/pgData/pgsql15/edbua_tbs02
92664	edbua_ixtbs01	/pgData/pgsql15/edbua_ixtbs01
92665	edbua_temptbs01	/pgData/pgsql15/edbua_temptbs01

(13 rows)

```
edbua=#
```

9. Move Table Between Tablespaces

```
edbua=# ALTER TABLE emp01 SET TABLESPACE edbua_tbs02;
ALTER TABLE
edbua=# ALTER TABLE emp02 SET TABLESPACE edbua_tbs02;
ALTER TABLE
edbua=# ALTER TABLE emp03 SET TABLESPACE edbua_tbs02;
ALTER TABLE
edbua=#
```

```
pg_tblspc/<tbs_oid>/PG_15_<catalog_version>/<db_oid>/<relfilen
ode>
```

```
edbua=# select pg_relation_filepath('emp01');
           pg_relation_filepath
-----
pg_tblspc/92663/PG_15_202209061/92666/92710
(1 row)
```

```
edbua=#
```



```

edbua=# SELECT relname, pg_relation_filepath(oid) AS filepath
FROM pg_class
WHERE relname IN
('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');
 relname | filepath
-----+-----
emp01    | pg_tblspc/92663/PG_15_202209061/92666/92710
emp02    | pg_tblspc/92663/PG_15_202209061/92666/92713
emp03    | pg_tblspc/92663/PG_15_202209061/92666/92716
emp01_ix | pg_tblspc/92664/PG_15_202209061/92666/92700
emp02_ix | pg_tblspc/92664/PG_15_202209061/92666/92701
emp03_ix | pg_tblspc/92664/PG_15_202209061/92666/92702
(6 rows)

```

edbua=#

Important: Indexes do not move automatically when you move a table.

They remain in their original tablespace (edbua_ixtbs01 in this case).

If you want indexes in edbua_tbs02, you must explicitly run:

```

edbua=# ALTER INDEX emp01_ix SET TABLESPACE edbua_tbs02;
ALTER INDEX
edbua=# ALTER INDEX emp02_ix SET TABLESPACE edbua_tbs02;
ALTER INDEX
edbua=# ALTER INDEX emp03_ix SET TABLESPACE edbua_tbs02;
ALTER INDEX
edbua=#
edbua=# SELECT c.relname, t.spcname AS tablespace,
pg_relation_filepath(c.oid)
edbua-# FROM pg_class c
edbua-# LEFT JOIN pg_tablespace t ON c.reltablespace = t.oid
edbua-# WHERE c.relname IN
('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');
 relname | tablespace | pg_relation_filepath
-----+-----+-----
emp01_ix | edbua_tbs02 |
pg_tblspc/92663/PG_15_202209061/92666/92719
emp02_ix | edbua_tbs02 |
pg_tblspc/92663/PG_15_202209061/92666/92720
emp03_ix | edbua_tbs02 |

```

```

pg_tblspc/92663/PG_15_202209061/92666/92721
emp01      | edbua_tbs02 |
pg_tblspc/92663/PG_15_202209061/92666/92710
emp02      | edbua_tbs02 |
pg_tblspc/92663/PG_15_202209061/92666/92713
emp03      | edbua_tbs02 |
pg_tblspc/92663/PG_15_202209061/92666/92716
(6 rows)

```

edbua=#

edbua=# SELECT

edbua=# c.relname,

edbua=# COALESCE(t1.spcname, t2.spcname) AS tablespace,

edbua=#

pg_catalog.pg_tablespace_location(COALESCE(c.reltablespace,
d.dattablespace)) AS tablespace_path,

edbua=# current_setting('data_directory') AS
data_directory,

edbua=# CASE

edbua=# WHEN c.reltablespace = 0 THEN

edbua=# current_setting('data_directory') ||
'/base/' || d.oid || '/' || c.relfilenode

edbua=# ELSE

edbua=#

pg_catalog.pg_tablespace_location(c.reltablespace) || '/PG_'
||

edbua=# split_part(split_part(version(), ' ', 2),
'.', 1) || '_' ||

edbua=# '202209061/' || d.oid || '/' ||
c.relfilenode

edbua=# END AS full_path_guess

edbua=# FROM pg_class c

edbua=# LEFT JOIN pg_tablespace t1 ON c.reltablespace = t1.oid

edbua=# LEFT JOIN pg_database d ON d.datname =

current_database()

edbua=# LEFT JOIN pg_tablespace t2 ON d.dattablespace = t2.oid

edbua=# WHERE c.relname IN

('emp01','emp02','emp03','emp01_ix','emp02_ix','emp03_ix');

relname | tablespace | tablespace_path |

data_directory | full_path_guess

```

-----+-----+-----+-----
-----+-----+-----+-----

```

```

-----
emp01_ix | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92719
emp02_ix | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92720
emp03_ix | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92721
emp01     | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92710
emp02     | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92713
emp03     | edbua_tbs02 | /pgData/pgsql15/edbua_tbs02 |
/pgData/pgsql15/data |
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92716
(6 rows)

```

edbua=#

10. Verify Database Directory (Filesystem)

```

edbua=# SELECT oid, datname FROM pg_database WHERE datname =
current_database();

```

```

   oid   | datname
-----+-----
 92666   | edbua
(1 row)

```

edbua=#

```

[root@pg17 ~]# ls -ltr
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/*
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92710
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92711
-rw-----. 1 postgres postgres 8192 Sep 23 00:14

```

```

/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92712
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92713
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92714
-rw-----. 1 postgres postgres 8192 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92715
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92716
-rw-----. 1 postgres postgres    0 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92717
-rw-----. 1 postgres postgres 8192 Sep 23 00:14
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92718
-rw-----. 1 postgres postgres 8192 Sep 23 00:52
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92719
-rw-----. 1 postgres postgres 8192 Sep 23 00:52
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92720
-rw-----. 1 postgres postgres 8192 Sep 23 00:52
/pgData/pgsql15/edbua_tbs02/PG_15_202209061/92666/92721
[root@pg17 ~]#

```

Issue: Why are there 12 files when I have only 6 objects?

created only: 3 tables: emp01, emp02, emp03 & **3 indexes:**
emp01_ix, emp02_ix, emp03_ix

**But in the tablespace directory, we're seeing 12 files: 3
tables + 3 indexes + 3 TOAST tables + 3 TOAST indexes**

```

edbua=# SELECT
edbua=#      c.relname,
edbua=#      c.relfilenode,
edbua=#      c.relkind,
edbua=#      n.nspname AS schema,
edbua=#      t.spcname AS tablespace
edbua=# FROM
edbua=#      pg_class c
edbua=# JOIN
edbua=#      pg_namespace n ON n.oid = c.relnamespace
edbua=# LEFT JOIN
edbua=#      pg_tablespace t ON c.reltablespace = t.oid
edbua=# WHERE

```

```

edbua=#      c.reltablespace = (SELECT oid FROM pg_tablespace
WHERE spcname = 'edbua_tbs02')
edbua=# ORDER BY
edbua=#      c.relfilenode;
      relname          | relfilenode | relkind | schema |
tablespace
-----+-----+-----+-----+-----
emp01                  |      92710 | r       | public |
edbua_tbs02
pg_toast_92685         |      92711 | t       | pg_toast |
edbua_tbs02
pg_toast_92685_index   |      92712 | i       | pg_toast |
edbua_tbs02
emp02                  |      92713 | r       | public |
edbua_tbs02
pg_toast_92690         |      92714 | t       | pg_toast |
edbua_tbs02
pg_toast_92690_index   |      92715 | i       | pg_toast |
edbua_tbs02
emp03                  |      92716 | r       | public |
edbua_tbs02
pg_toast_92695         |      92717 | t       | pg_toast |
edbua_tbs02
pg_toast_92695_index   |      92718 | i       | pg_toast |
edbua_tbs02
emp01_ix               |      92719 | i       | public |
edbua_tbs02
emp02_ix               |      92720 | i       | public |
edbua_tbs02
emp03_ix               |      92721 | i       | public |
edbua_tbs02
(12 rows)

edbua=#

```

11. Rename tablespace

```

edbua=# ALTER TABLESPACE edbua_tbs02 RENAME TO edbua_tbs03;
ALTER TABLESPACE
edbua=#

```

```
edbua=# \db+ edbua*
```

List of

tablespaces

Name	Owner	Location	
Access privileges	Options	Size	Description
edbua_ixtbs01	postgres	/pgData/pgsql15/edbua_ixtbs01	
		6 bytes	
edbua_tbs01	postgres	/pgData/pgsql15/edbua_tbs01	
		7789 kB	
edbua_tbs03	postgres	/pgData/pgsql15/edbua_tbs02	
		48 kB	
edbua_temptbs01	postgres	/pgData/pgsql15/edbua_temptbs01	
		6 bytes	

(4 rows)

```
edbua=#
```

12. Drop Tablespace

```
edbua=# drop tablespace edbua_tbs03;
ERROR:  tablespace "edbua_tbs03" is not empty
edbua=#
```

```
edbua=# SELECT
edbua-# CASE
edbua-#     WHEN c.relkind = 'r' THEN
edbua-#         format('ALTER TABLE %I.%I SET TABLESPACE
edbua_tbs01;', n.nspname, c.relname)
edbua-#     WHEN c.relkind = 'i' THEN
edbua-#         format('ALTER INDEX %I.%I SET TABLESPACE
edbua_tbs01;', n.nspname, c.relname)
edbua-#     WHEN c.relkind = 'S' THEN
edbua-#         format('ALTER SEQUENCE %I.%I SET TABLESPACE
edbua_tbs01;', n.nspname, c.relname)
edbua-#     END AS alter_statement
edbua-# FROM pg_class c
edbua-# JOIN pg_namespace n ON n.oid = c.relnamespace
edbua-# WHERE c.reltablespace = (SELECT oid FROM pg_tablespace
```

```

WHERE spcname = 'edbua_tbs03')
edbua=# AND c.relkind IN ('r', 'i', 'S')
edbua=# AND n.nspname NOT IN ('pg_catalog', 'pg_toast',
'information_schema')
edbua=# ORDER BY n.nspname, c.relname;
          alter_statement
-----

```

```

ALTER TABLE public.emp01 SET TABLESPACE edbua_tbs01;
ALTER INDEX public.emp01_ix SET TABLESPACE edbua_tbs01;
ALTER TABLE public.emp02 SET TABLESPACE edbua_tbs01;
ALTER INDEX public.emp02_ix SET TABLESPACE edbua_tbs01;
ALTER TABLE public.emp03 SET TABLESPACE edbua_tbs01;
ALTER INDEX public.emp03_ix SET TABLESPACE edbua_tbs01;
(6 rows)

```

edbua=#

```

edbua=# ALTER TABLE public.emp01 SET TABLESPACE edbua_tbs01;
ALTER INDEX public.emp02_ix SET TABLESPACE edbua_tbs01;
ALTER TABLE public.emp03 SET TABLESPACE edbua_tbs01;
ALTER INDEX public.emp03_ix SET TABLESPACE edbua_tbs01;ALTER
TABLE

```

```

edbua=# ALTER INDEX public.emp01_ix SET TABLESPACE
edbua_tbs01;
ALTER INDEX

```

```

edbua=# ALTER TABLE public.emp02 SET TABLESPACE edbua_tbs01;
ALTER TABLE

```

```

edbua=# ALTER INDEX public.emp02_ix SET TABLESPACE
edbua_tbs01;
ALTER INDEX

```

```

edbua=# ALTER TABLE public.emp03 SET TABLESPACE edbua_tbs01;
ALTER TABLE

```

```

edbua=# ALTER INDEX public.emp03_ix SET TABLESPACE
edbua_tbs01;
ALTER INDEX
edbua=#

```

```

edbua=# drop tablespace edbua_tbs03;
DROP TABLESPACE
edbua=# \db+ edbua*

```

List of

tablespaces

Name	Owner	Location
------	-------	----------

```

| Access privileges | Options | Size | Description
-----+-----+-----+-----
+-----+-----+-----+-----
edbua_ixtbs01 | postgres | /pgData/pgsql15/edbua_ixtbs01
| | | 6 bytes |
edbua_tbs01 | postgres | /pgData/pgsql15/edbua_tbs01
| | | 7837 kB |
edbua_temptbs01 | postgres | /pgData/pgsql15/edbua_temptbs01
| | | 6 bytes |
(3 rows)

edbua=#

```

13. Change ownership for Tablespace

```

edbua=# ALTER TABLESPACE edbua_tbs01 OWNER TO teja;
ALTER TABLESPACE
edbua=# \db+ edbua*

```

List of

tablespaces

```

      Name      | Owner  | Location
| Access privileges | Options | Size | Description
-----+-----+-----+-----
+-----+-----+-----+-----
edbua_ixtbs01 | postgres | /pgData/pgsql15/edbua_ixtbs01
| | | 6 bytes |
edbua_tbs01 | teja | /pgData/pgsql15/edbua_tbs01
| | | 7837 kB |
edbua_temptbs01 | postgres | /pgData/pgsql15/edbua_temptbs01
| | | 6 bytes |
(3 rows)

```

```

edbua=# ALTER TABLESPACE edbua_tbs01 OWNER TO postgres;
ALTER TABLESPACE
edbua=# \db+ edbua*

```

List of

tablespaces

```

      Name      | Owner  | Location
| Access privileges | Options | Size | Description
-----+-----+-----+-----
+-----+-----+-----+-----

```



```

edbua_ixtbs01      | postgres | /pgData/pgsql15/edbua_ixtbs01
|                  |          | 6 bytes |
edbua_tbs01        | postgres | /pgData/pgsql15/edbua_tbs01
|                  |          | 7837 kB |
edbua_temptbs01    | postgres | /pgData/pgsql15/edbua_temptbs01
|                  |          | 6 bytes |
(3 rows)

```

edbua=#

14. Verify Symbolic Links

```
edbua=# show data_directory;
```

```
data_directory
```

```
-----
```

```
/pgData/pgsql15/data <----
```

```
(1 row)
```

edbua=#

```
[root@pg17 pg_tblspc]# ls -ltr
```

```
/pgData/pgsql15/data/pg_tblspc/*
```

```
lrwxrwxrwx. 1 postgres postgres 22 Aug 18 06:36
```

```
/pgData/pgsql15/data/pg_tblspc/24581 -> /u01/data/dell_ts_data
```

```
lrwxrwxrwx. 1 postgres postgres 23 Aug 18 06:36
```

```
/pgData/pgsql15/data/pg_tblspc/24582 ->
```

```
/u01/data/dell_ts_index
```

```
lrwxrwxrwx. 1 postgres postgres 20 Sep 3 09:52
```

```
/pgData/pgsql15/data/pg_tblspc/24594 -> /pgData/pgsql15/data
```

```
lrwxrwxrwx. 1 postgres postgres 29 Sep 3 11:55
```

```
/pgData/pgsql15/data/pg_tblspc/24597 ->
```

```
/pgData/pgsql15/data/trd_tb01
```

```
lrwxrwxrwx. 1 postgres postgres 29 Sep 3 11:55
```

```
/pgData/pgsql15/data/pg_tblspc/24598 ->
```

```
/pgData/pgsql15/data/trd_ix01
```

```
lrwxrwxrwx. 1 postgres postgres 27 Sep 4 05:59
```

```
/pgData/pgsql15/data/pg_tblspc/32787 ->
```

```
/pgData/pgsql15/pgtb/geb_tb
```

```
lrwxrwxrwx. 1 postgres postgres 27 Sep 4 06:03
```

```
/pgData/pgsql15/data/pg_tblspc/32788 ->
```

```
/pgData/pgsql15/pgix/geb_ix
```

```
lrwxrwxrwx. 1 postgres postgres 27 Sep 20 07:20
/pgData/pgsql15/data/pg_tblspc/92662 ->
/pgData/pgsql15/edbua_tbs01
lrwxrwxrwx. 1 postgres postgres 29 Sep 20 07:21
/pgData/pgsql15/data/pg_tblspc/92664 ->
/pgData/pgsql15/edbua_ixtbs01
lrwxrwxrwx. 1 postgres postgres 31 Sep 20 07:21
/pgData/pgsql15/data/pg_tblspc/92665 ->
/pgData/pgsql15/edbua_temptbs01
[root@pg17 pg_tblspc]#
```

15. How to find tablespace size

```
edbua=# SELECT spcname,
pg_size_pretty(pg_tablespace_size(spcname)) AS size FROM
pg_tablespace;
```

spcname	size
pg_default	1660 MB
pg_global	571 kB
dell_ts_data	7761 kB
dell_ts_index	16 kB
dell_tbs	0 bytes
trd_tb01	1615 MB
trd_ix01	2648 kB
geb_tb	13 GB
geb_ix	8950 MB
edbua_ixtbs01	6 bytes
edbua_temptbs01	6 bytes
edbua_tbs01	7837 kB

(12 rows)

```
edbua=#
```

Caution: Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.

Thank you,

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